

UEBAI (Universal Educational Bots): Virtual AI Robot Teachers That Read Topic Cards, Not the Whole Textbook

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Introduction

More and more students ask AI chatbots about their school textbooks. Usually you upload the whole book, and the AI reads hundreds of pages again (100,000 to 190,000 tokens) for **every single question**. That makes answers slow and expensive, only big AI models can handle it, and research shows that models miss facts buried in the middle of long texts (Liu et al.). The common fix, called RAG, cuts the book into equal pieces, but a piece often stops in the middle of an idea, and it doesn't know that the textbook and the teacher's slides are talking about the same topic.

I built UEBAI, an Arabic-first web app where students design their own 3D robot teacher and give it their study materials. UEBAI reads the book **once** and turns it into topic cards. After that, each question is answered from the right card, and every page number the teacher mentions is checked. My goals were to build a working prototype, test it against reading the whole book on real Saudi Ministry of Education textbooks (with answer keys I wrote before any test), and see what happens when a second source is added.

Research question: Can an AI teacher that reads only one topic card answer as well as one that reads the whole book, and do it faster and cheaper?

THE PROBLEM

1 The whole book, every time
about 160,000 tokens per question

2 Slow and expensive
long waits, only big models

3 Many sources
slides and book overlap or disagree

HOW UEBAI DOES IT

Read the book once
and turn it into topic cards

Open one card
about 3,000 tokens per question

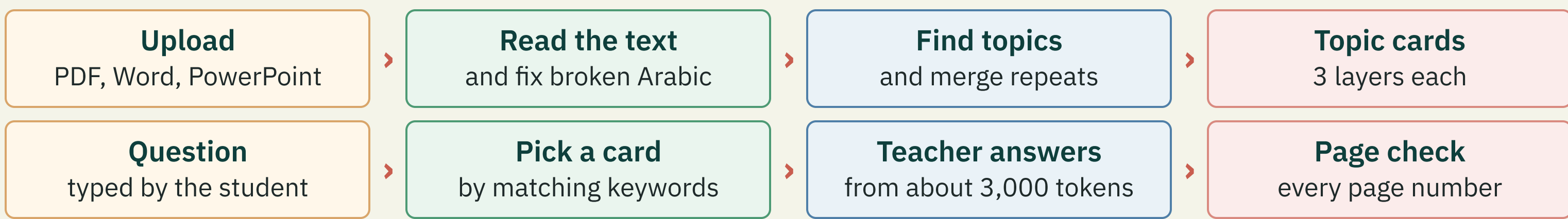
Check the pages
made-up page numbers are removed

Topic cards • Merged sources • Checked pages • Arabic first

Fig. 1. Reading the whole book for every question is slow and expensive. UEBAI reads the book once, then opens only the card each question needs.

Methodology

1 How it works



2 Test plan

1 Two MoE books Biology 1 and History	2 Answer keys first 74 questions, random pages	3 5 Gemini models same instructions	4 Paired tests whole book vs UEBAI
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3 Measurements

Setup:	Each question was asked twice on the same model, once with the whole book and once with UEBAI, in random order. 27 questions on up to 5 models gave 69 pairs. 7 questions were not in the book on purpose.
Measures:	Tokens read, time until the first word, total time, key facts in the answer (checked by code), the right page, and whether it says so when something isn't in the book.
Statistics:	Paired permutation test and 95% confidence intervals. Before testing, I decided UEBAI counts as "not worse" if it loses less than 10 points on key facts.
Fairness:	I wrote the answer keys before any test and saved their SHA-256 fingerprint, so they can't be changed later. A free extra check covered all 66 answerable questions.

Results

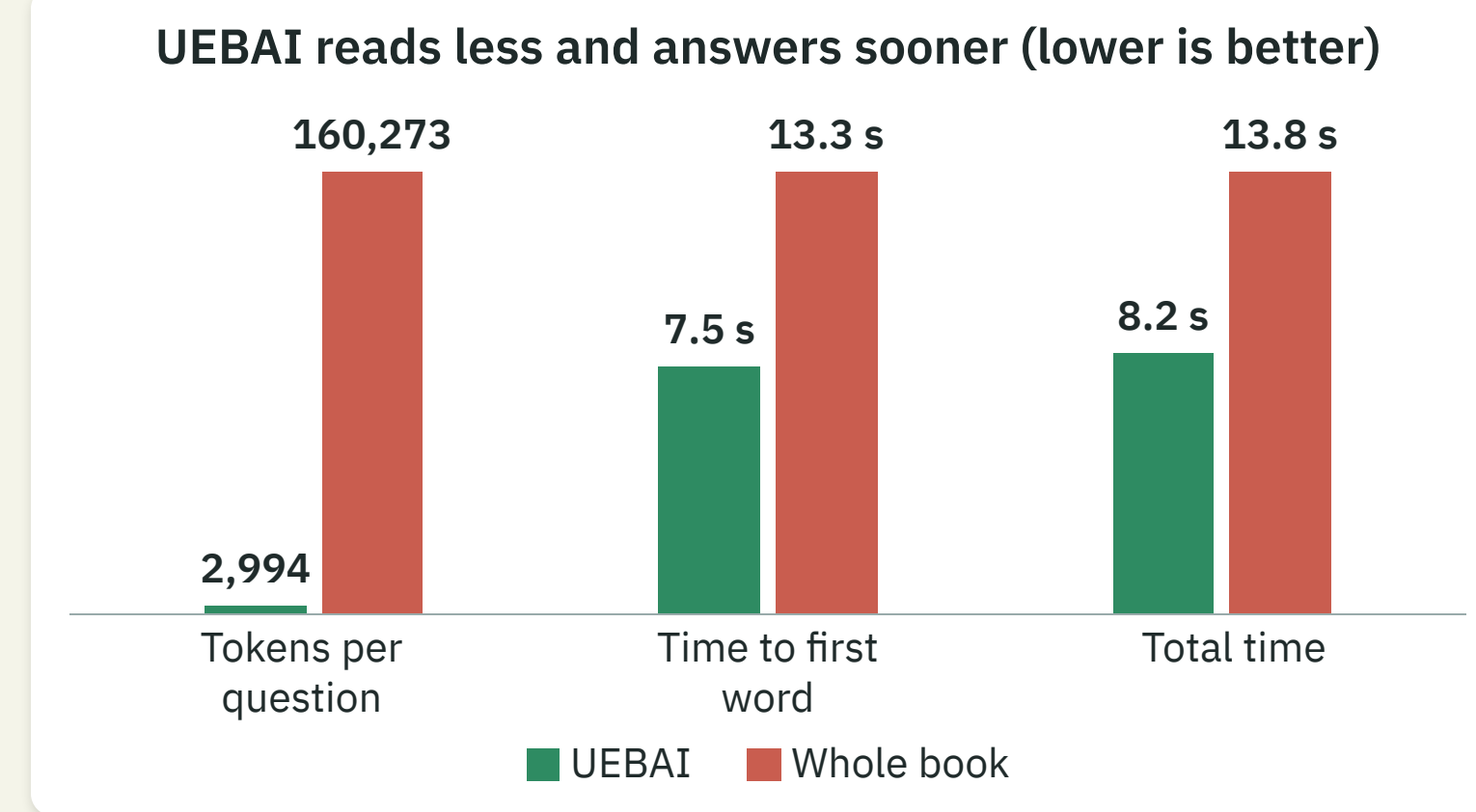
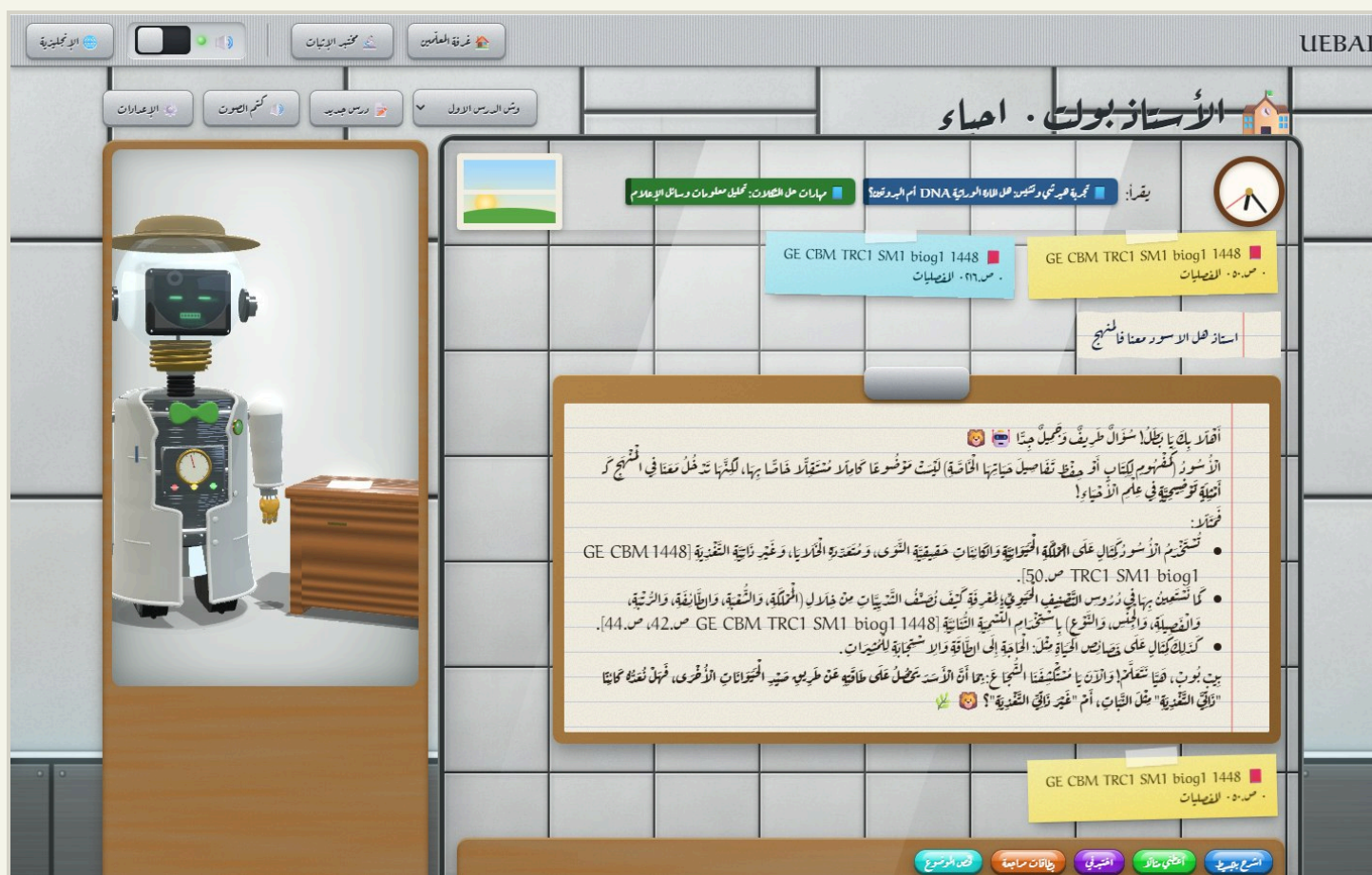


Fig. 2. The working prototype: an answer with page numbers, and the cards the teacher read.

55× fewer tokens	2× faster first word	Not worse in answer quality	1.7% of the book read
Measure (69 pairs)	UEBAI	Whole book	Result
Tokens per question	2,994	160,273	55× fewer in 69/69, $p < 0.0001$
Time to first word	7.5 s	13.3 s	faster in 49/69, $p = 0.024$
Total time	8.2 s	13.8 s	faster in 48/69, $p = 0.037$
Key facts in the answer	92%	97%	not worse (CI −9 to +1 points)
Right page cited	61/63	63/64	about the same

Times are averages, and the typical (median) speed-up was 2×. The gain depends a lot on the model: in the first biology test it went from 28× on gemini-3-flash-preview down to only 1.1× on gemini-2.5-flash. In the free check, UEBAI's cards held 95% of the biology answer facts while being just 1.7% of the book.

Innovation

UEBAI organizes the book by **study topic** instead of cutting it into equal pieces. The idea builds on RAG (Lewis et al.). What's new is how I designed and combined the parts for school textbooks.

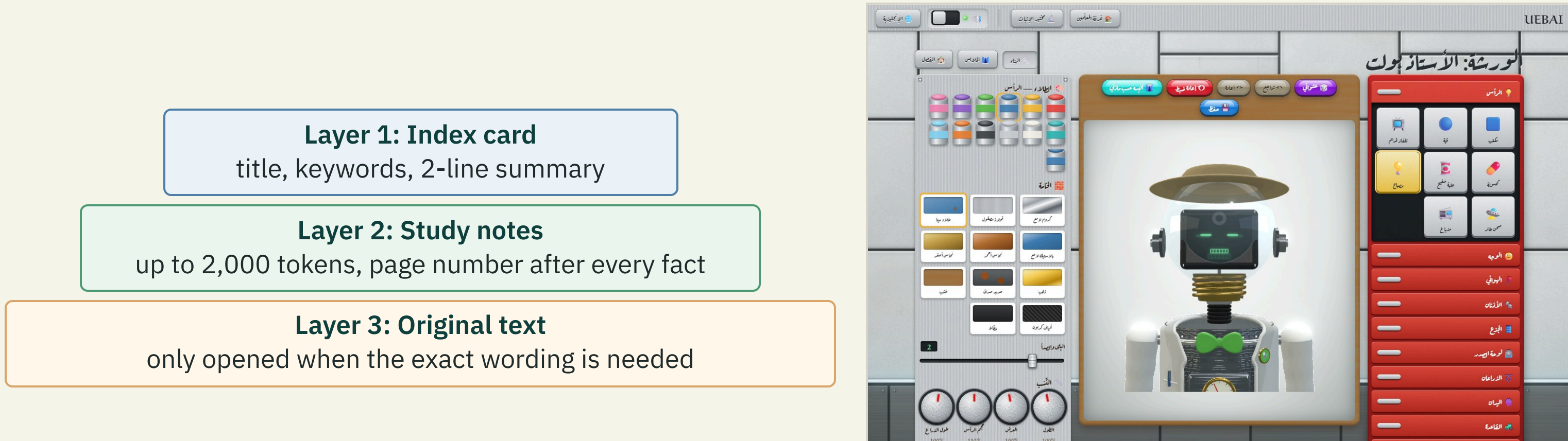


Fig. 3. Students design their own 3D teacher.

Merging sources	When I added the teacher's slides, 27 topics joined existing book cards, and UEBAI flagged 2 places where the sources disagree (fungi: 4 or 5 phyla?)
Checked page numbers	Every page the teacher mentions is checked against the book
Cards teachers can fix	Teachers can edit, merge and split cards in the library

Conclusion

Main conclusion

Reading one topic card instead of the whole book cut the text per question 55 times and made answers start about twice as fast, and the answers were statistically not worse.

Performance 92% of key facts vs 97%, over 69 paired tests	Impact about 55× less text to pay for, and small models can handle it	Readiness Working prototype tested on 2 MoE textbooks
What I predicted before testing	What happened	
H1 Faster: median at least 3×	Partly	2× overall ($p = 0.024$), 3× or more on 2 of 4 models
H2 Cheaper: at least 20× fewer tokens	Yes	55× fewer, in 69 of 69 pairs
H3 Quality drops less than 10 points	Yes	for both books together; History alone needs more tests

Limitations

The speed gain depends on the model (1.1× to 28×). In History, keyword matching sometimes opens the wrong card, because the same people appear in many units (85% vs 87% for plain chunks). There's no classroom trial or blind teacher rating yet; a biology teacher gave it 9/10, but informally. I also expected the extra slides to hurt plain chunks. They didn't, and I'm reporting that.

Future Work & References

- Blind rating by teachers, and a trial with real students in class.
- Smarter card picking that opens more cards when topics are close.
- Student and teacher accounts, and connecting to school platforms.

- Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS.
- Liu, N. F. et al. (2024). Lost in the Middle: How Language Models Use Long Contexts. TACL.
- Sarathi, P. et al. (2024). RAPTOR: Recursive Abstractive Processing for Tree-Organized Retrieval. ICLR.
- Edge, D. et al. (2024). From Local to Global: A Graph RAG Approach to Query-Focused Summarization. arXiv:2404.16130.
- Saudi Ministry of Education. Biology 1 and History secondary textbooks.

UEBAI PROJECT FILES



github.com/RashedAhmedAseri/UEBAI